python™

Python Track

Time and Space Complexity

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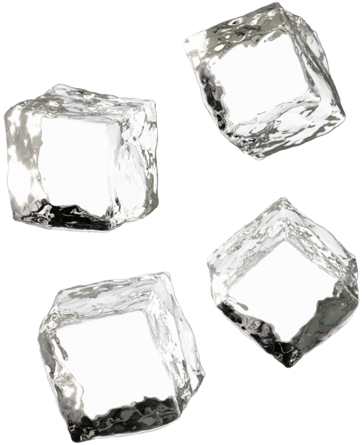

What is time and space complexity?

Time complexity

It measures an algorithm’s execution time with respect to input size.

Space complexity

- The space Complexity of an algorithm is the total auxiliary(extra) space taken by the algorithm with respect to the input size.
- An in-place algorithm is one that **does not** use extra space (or uses only O(1) auxiliary space) beyond the input itself. Instead, it modifies the input It self.

Cont...

- Input space: is the expected space.
- Auxiliary space: The additional space used by the algorithm, e.g., to hold temporary variables or the space used by the call stack.

Asymptotic Analysis

- Asymptotic analysis evaluates the performance of algorithms as input size grows, crucial for predicting efficiency and resource usage.

- Focuses on limiting behavior of an algorithm, disregarding hardware and software factors.

- Enables comparison of worst, average, and best-case scenarios across different algorithms.

What are the different cases in time and space complexity analysis?

- Best Case
- Worst Case
- Average Case
- Amortized Case

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Best case - (Big Ω)

- In the best-case analysis, we calculate the lower bound on the running time of an algorithm.

- We must know the case that causes a minimum number of operations to be executed.

Average case

- In average case analysis, we take all possible inputs and calculate the computing time for all of the inputs.
- Sum all the calculated values and divide the sum by the total number of inputs
- Worst-case inputs may rarely occur in practice, making average-case analysis more practical.

Worst case- (Big O)

- In the worst-case analysis, we calculate the upper bound on the running time of an algorithm.
- We must know the case that causes a maximum number of operations to be executed.
- As Competitive Programmers, we care about worst case complexities.

Example 1 - Linear search

- Best case: the best case occurs when x is present at the first location.

- Worst case: the worst case happens when the element to be searched (x) is not present in the array.

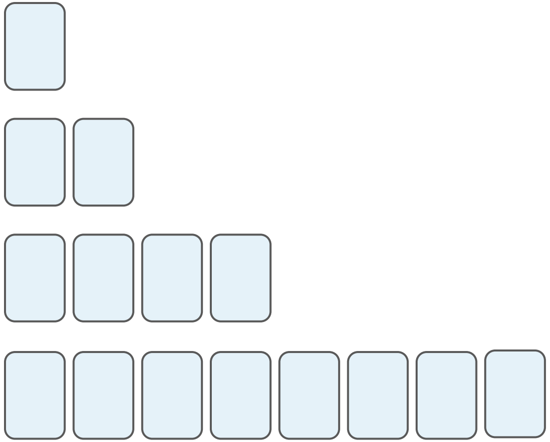
Amortized case

- Amortized time is the way to express the time complexity when an algorithm has the very bad time complexity only once in a while besides the time complexity that happens most of time.

Amortized case

Example 2: Consider a dynamic array that doubles in size when it reaches its capacity:

- Most insertions (append) take O(1) time.
- Once it is full, it will copy the values to another space with 2n space capacity (O(n) operation).



Basic Steps to calculate time complexity

Identify the basic operations

- Assignment
- Comparison
- Arithmetic operations
- Array element access
- Conditional Statements
- Loop operations
- Function calls
- Logical operations
- Bitwise operations

=> When analyzing an algorithm, go through the code and identify these basic operations

Count the operations

- Analyze loops and iterations
- Consider conditional statements
- Look for recursion
- Account for nested operations
- Count function calls

Analyze loops and iterations

```python
for i in range(n):
    # code

for i in range(3 * n):
    # code

for i in range(3 + n):
    # code
```

```python
mx = float('-inf')
mn = float('inf')

for x in array:
    mx = max(mx, x)

for x in array:
    mn = min(mx, x)
```

```python
mx = float('-inf')
mn = float('inf')

for x in array:
    mx  = max(mx, x)
    mn  = min(mx, x)
```

Consider conditional statements

Assuming `is_even(i)` is constant time operation

```
for i in nums:
    if is_even(i):
        for j in nums:
            # constant time code here
```

Look for recursion

```python
def factorial(n):
    if n == 0 or n == 1:
        return 1
    else:
        return n * factorial(n-1)

# Example usage
result = factorial(5)
print(result)
```

```python
def Fibonacci(n):
    if n == 0:
        return 0
    elif n == 1 or n == 2:
        return 1
    else:
        return Fibonacci(n-1) +
Fibonacci(n-2)

# Example usage
result = Fibonacci(9)
print(result)
```

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Account for nested operations

```python
for i in range(n):
    for j in range(i, n):
        # code
```

```python
def example_code(arr):
    n = len(arr)
    result = 0
    for element in arr:
        i = 1
        while i * i <= n:
            result += element
            i += 1
    return result
```

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Count function calls

```
def example_function_1(x):
    return x*2

def example_function_2(x):
    total = 0
    for element in arr:
        total += example_function_1(element)
    return total

def main_algorithm(arr):
    result = 0
    for _ in range(3):
        result += example_function_2(arr)
    return result
```

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Express the relationship

- Express the total number of operations as a mathematical expression of the input size ‘n’.

Simplify to Big O Notation

- Drop the non-dominant terms
- Drop the constants

Drop the non-dominant Terms

- We should drop the non-dominant terms
- Hence,
  - $O(N^2 + N) = O(N^2)$
  - $O(N + \log N) = O(N)$
- We might still have a sum in the runtime.
  - $O(B^2 + A)$ can not be reduced
  - without some special knowledge of A and B.

Drop the Constants

- We drop the constants in run time

- An algorithm that one might have described as O(2N) is actually O(N).

```python
mx = float('-inf')
mn = float('inf')

for x in array:
    mx = max(mx, x)

for x in array:
    mn = min(mx, x)
```

```python
mx = float('-inf')
mn = float('inf')

for x in array:
    mx  = max(mx, x)
    mn  = min(mx, x)
```

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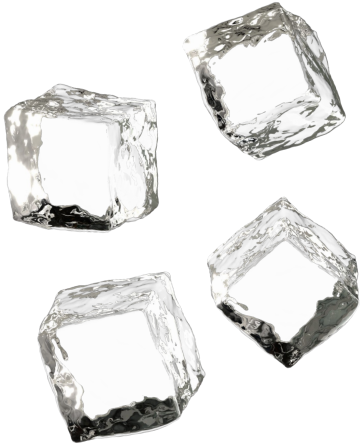
General Cases

```python
n, i = 100, 1
while i <= n:
    i *= 2

while n != 0:
    n //= 2
```

Anytime we are multiplying or dividing our loop variable by some constant other than 1, we are dealing with logarithmic time complexities.

Logarithmic time is very good; for a given dataset of maximum length 5 * 10 ^ 9, an nlogn solution is the same as 32 * n.



How to calculate space complexity?

O(1) - Constant Space

```python
def getMin(arr):
    min_num = float('inf')
    N = len(arr)
    for i in range(N):
        min_num = min(min_num, arr[i])
return min_num
```

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O(m) - Linear Space

```
n = len(list_1)
m = len(list_2)

for i in range(n):
    temp = []
    for j in range(m):
        temp.append(i*j)
    print(temp)
```

Why is it not O(n x m)?

O(n*m) - Quadratic space

```
n = len(list_1)
m = len(list_2)

grid = []
for i in range(n): # n opperatio
    temp = [] n opperatio
    for j in range(m): #n
        temp.append(0)  #n * m
    grid.append(temp) # n

Total = n * m
```

O(c) -> O(1)

```python
# s is string
freq = {}
max_freq = 0
for i in range(len(s)):
    if s[i] in freq:
        freq[s[i]] += 1
    else:
        freq[s[i]] = 1
    max_freq = max(max_freq, freq[s[i]])
print(max_freq)
```

Why is it O(1)?

Time complexities of common data structures

Data Structure	Access	Search	Insertion	Deletion
List	O(1)	O(n)	O(1)*	O(n)
Tuple	O(1)	O(n)	N/A	N/A
Set	N/A	O(1)*	O(1)*	O(1)*
Dictionary	O(1)*	O(1)*	O(1)*	O(1)*
Stack (via list)	O(n)	O(n)	O(1)	O(1)
Queue (via deque)	O(n)	O(n)	O(1)	O(1)
Linked List	O(n)	O(n)	O(1)	O(1)
Binary Search Tree	O(log n)*	O(log n)*	O(log n)*	O(log n)*
Heap	O(n)	O(n)	O(log n)	O(log n)

Note:

List: The amortized time for insertion is O(1) when adding to the end of the list. Insertions or deletions at arbitrary positions require shifting elements, making it O(n).

Set and Dictionary: Average-case complexities are O(1), but these can degrade to O(n) in the worst case due to hash collisions.

Binary Search Tree: For a balanced tree, like AVL or Red-Black Tree, operations are O(log n). In an unbalanced tree, they can degrade to O(n).

Tuples: Tuples are immutable, so they do not support insertion or deletion.

Tips and Pitfalls

Tips - Common time complexities

n: input size of the problem	Acceptable time complexity
n ≤ 10	O(n!)
n ≤ 20	O(2^n)
n ≤ 30	O(n^4)
n ≤ 100	O(n^3)
n ≤ 10^3	O(n^2)
n ≤ 10^5	O(n log n)
n ≤ 10^6	O(n)
n > 10^8	O(log n) or O(1)

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Tips - Examples

- O(n!) [Factorial time]: Permutations of 1 ... n
- O(2^n) [Exponential time]: Exhaust all subsets of an array of size n
- O(n^3) [Cubic time]: Exhaust all triangles with side length less than n
- O(n^2) [Quadratic time]: Slow comparison-based sorting (eg. Bubble Sort, Insertion Sort, Selection Sort)
- O(n log n) [Linearithmic time]: Fast comparison-based sorting (eg. Merge Sort)
- O(n) [Linear time]: Linear Search (Finding maximum/minimum element in a 1D array), Counting Sort
- O(log n) [Logarithmic time]: Binary Search, finding GCD (Greatest Common Divisor) using Euclidean Algorithm
- O(1) [Constant time]: Calculation (eg. Solving linear equations in one unknown)

Tips - Space Usage

Data Type	Size (Approximate)
None	16 bytes
int	28 bytes (small integers), increases with size
float	24 bytes
str	49 bytes (overhead) + 1 byte per character
list	64 bytes (empty list), 8 bytes per elements reference
tuple	48 bytes (empty tuple), 8 bytes per elements reference
dict	240 bytes (empty dict), 8 bytes per key-value reference
set	224 bytes (empty set), 8 bytes per elements reference

Tips - What does this mean?

It depends on the processor involved. For a processor running 4GHz, it would mean 4*10^9 cycles per-second. If each operation take one cycle, it means the computer can run 10^9 operations per second. (time complexity should be less than 10^9)

time limit per test: 2 seconds
memory limit per test: 256 megabytes
input: standard input
output: standard output

Tips - Common Errors

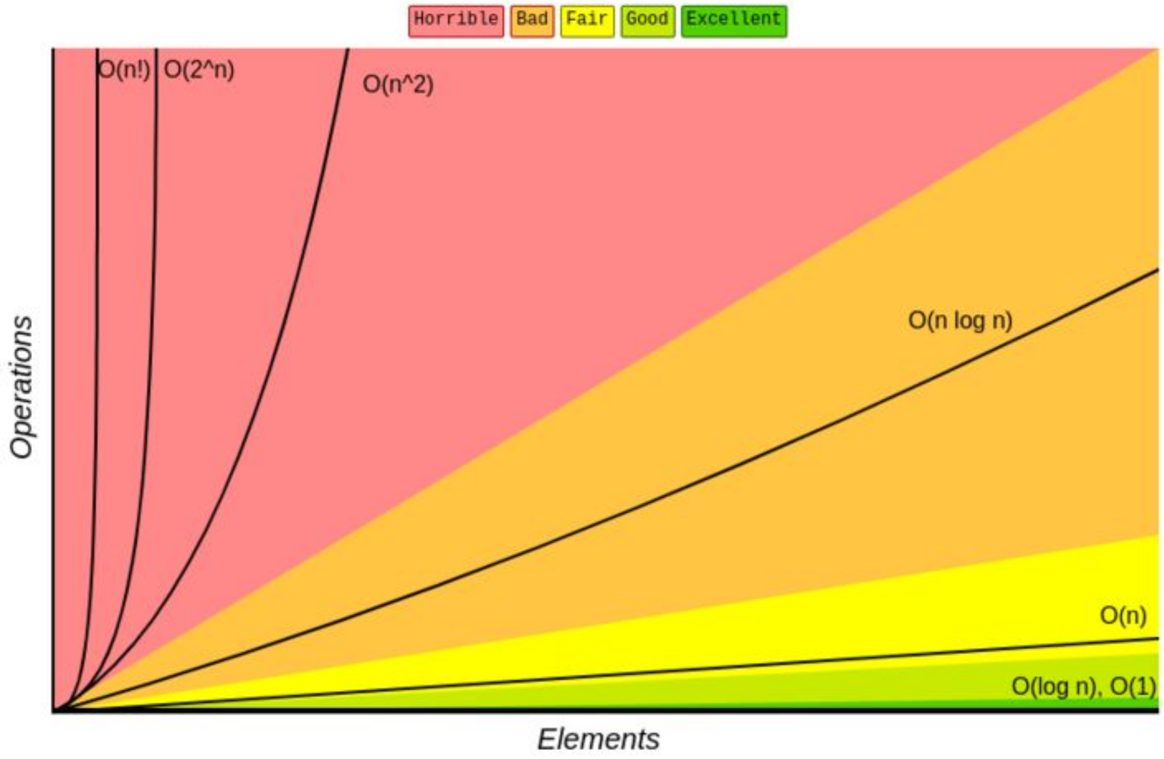
Time Limit Exceeded: The program hadn't terminated in time indicated in the problem statement

Memory Limit Exceeded: The program tries to consume more memory than is indicated in the problem statement

Runtime Error (): The program terminated with a non-zero return code (possible reasons: array out of bound error, division by zero, stack overflow, incorrect pointers usage, recursion limit exceeded, etc)

Compilation Error (CE): The program failed to compile due to syntax errors, missing libraries, type mismatches, or other coding mistakes that prevent successful compilation.

Tips



Horrible
Bad
Fair
Good
Excellent
O(n!)
O(2^n)
O(n^2)
O(n log n)
O(n)
O(log n), O(1)
Operations
Elements

Pitfalls

- Don't use the same input size notation for different inputs

```
for i in range(len(input_array)):
        for j in range(max_value):
            total_operations += 1
```

Time Complexity : O(n * m), where:

- n = length of input_array
- m = max_value

Pitfalls

- Don't include the input space in your space analysis; focus on the auxiliary space unless specified otherwise.

  nums.sort() => Space complexity : O(1)
  sorted(nums) => Space Complexity : O(n)

- Expected memory is not considered when calculating space complexity. E.g If a question asks to build a list and return it, it doesn't count.

Pitfalls

- Don’t forget Big-Oh is an expression for the upper bound of an algorithm’s performance. It is possible for an algorithm’s performance to have different upper bounds.

- Note: Always pick the upper bound that does not underestimate your algorithm’s performance in your analysis.

Resources

- How to determine the solution of a problem by looking at its constraints? - Codeforces
- A Time Complexity Guide - Codeforces
- Asymptotic notation (article) | Algorithms | Khan Academy
- Codeforces Contest Rules
- Gayle Laakmann McDowell - Cracking the Coding Interview_ 189 Programming Questions and Solutions-CareerCup (2015)

Quote of the Day

“The first principle is that you must not fool yourself — and you are the easiest person to fool.”

- Richard Feynman